CHEMCOMM
Communication
www.rsc.org/chemcomm

The first zinc-promoted, environmentally friendly, and highly efficient acetoxyallylation of aldehydes in aqueous ammonium chloride

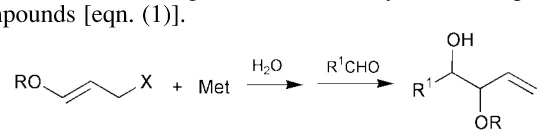
Marco Lombardo,* Rugiada Girotti, Stefano Morganti and Claudio Trombini*

Dipartimento di Chimica 'G.Ciamician', University of Bologna, Bologna, Italy. E-mail: trombini@ciam.unibo.it; Fax: +39 051 2099456; Tel: +39 051 2099513

Received (in Cambridge, UK) 9th August 2001, Accepted 20th September 2001
First published as an Advance Article on the web 15th October 2001

An exceptionally mild acetoxyallylation of aldehydes in water promoted by zinc is reported, using 3-bromo-1-acetoxyprop-1-ene as starting material; simple diastereoselectivity mainly depends on the nature of the aldehyde.

In any synthetic plan, carbon–carbon bond forming reactions are the most important and valuable transformations. If the construction of a carbon–carbon bond is carried out in water at ambient temperature, we also meet some basic requirements of green chemistry,[1] such as preventing hazardous waste (organic solvents), optimising synthetic efficiency by reducing derivatisation reactions (protection–deprotection *etc.*) and minimising energy consumption. The Henry nitroaldol reaction[2] has been known for a long time as a prototype of these reactions, allowing for example the direct use of unprotected aldoses,[3] but only in the 1980s did the use of water in organic chemistry enjoy an outbreak in terms of studies and applications.[4] A lot of attention has been devoted to the allylation of carbonyl compounds in water,[5] by generating allylic organometallic reagents, particularly tin, zinc, indium and even magnesium[6] species, and reacting them with carbonyl compounds. Two general protocols are available: the *in situ* Barbier procedure where the allylic species is formed in the presence of the carbonyl compound, or the stepwise Grignard procedure where the organometallic compound is preformed and then added to the aldehyde. Again, sugars may be used without derivatisation, as demonstrated by the synthesis of higher sugars,[7] sialic acid derivatives,[8] and *C*-branched sugars in water.[9] To the best of our knowledge, no report is available in the literature on Barbier reactions in water using 3-halo-1-alkoxyprop-1-enes **1**, which could ensure a straightforward access to 3-alkoxy-1-en-4-ols **2**,[10] attractive building blocks for the synthesis of polyoxo compounds [eqn. (1)].

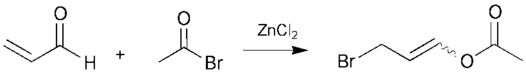


(1)

1 **2**

A possible reason why chemists are discouraged from attempting this reaction could be based on the sensitivity towards acidic conditions of the enol ether functionality in **1**. In fact, both Barbier protocols involving zinc and indium operate in acidic media; zinc requires the NH_4Cl–H_2O system to give the best results, while indium salts formed during the oxidative addition undergo acidic hydrolysis which lowers the pH by up to 3.

Considering that enol esters are supposed to be somewhat more stable than enol ethers under acidic conditions, we envisaged 3-bromo-1-acetoxyprop-1-ene (**3**) as a promising candidate for the acetoxyallylation of aldehydes. Preparation of **3** as a 35:65 *E*/*Z* isomer mixture,† was carried out by the $ZnCl_2$ catalysed 1,4-bromoacetylation of propenal with acetyl bromide [eqn. (2)].[11]

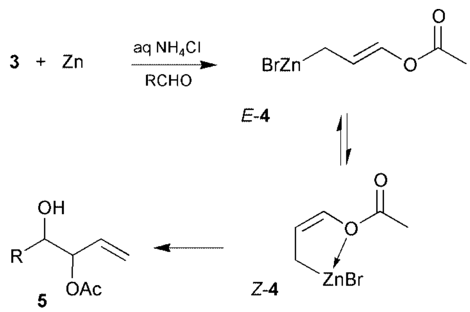


(2)

3

We were delighted by the observation that **3** was compatible with aq. ammonium chloride solutions and that the reaction of **3** with zinc was very rapid and exothermic. On this basis, a simple, fast and environmentally benign Barbier protocol was developed (Scheme 1);‡ preliminary results are collected in Table 1.

As shown in the reaction mechanism, once **4** is formed in water at pH = 5, a few reaction channels are possible: *E*/*Z* metallotropic isomerisation, protonation, Wurtz-like dimerisation, addition to an electrophile, if present. In the presence of an aldehyde, nucleophilic addition to the carbonyl group is faster than protonation or dimerisation. Of course, in the absence of the aldehyde, **4** completely decomposes within 30 min. Data presented in Table 1 deserve a few comments: i) with reference to the general economy criteria, the overall procedure is simple, fast and makes use of inexpensive reagents; ii) the procedure is based on **3**, an easily available starting material obtained in 70% yield in multigram scale, iii) side-products are limited to some Wurtz coupling product (< 10% based on **3**), and iv) the product is a selectively monoprotected diol, namely a 3-acetoxy-1-en-4-ol **5**, a straightforward precursor of the parent 1-en-3,4-diol by simple hydrolysis (quantitative in 2 h using K_2CO_3 in MeOH–H_2O 9:1). All these features make **3** highly competitive with alkoxylated organometallic derivatives **6a–f**,[12] whose forma-



Scheme 1

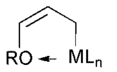
Table 1 Zinc-mediated acetoxyallylation of aldehydes under Barbier conditions in aq. NH_4Cl

Run	Aldehyde	*T* (°C)[a]	**5** Yield (%)	*syn*:*anti*
1	Benzaldehyde	22	90 (75)[b]	65:35 (70:30)[b]
2	2-Furaldehyde	2	81	80:20
3	*p*-Tolualdehyde	24	60[c]	60:40
4	*p*-Anisaldehyde	24	72[c]	70:30
5	(*E*)-PhCH = CHCHO	20	81	60:40
6	*n*-C_6H_{13}CHO	22	84	30:70
7	*n*-$C_{11}H_{23}$CHO	20	42	25:75
8	$PhCH_2CH_2CHO$	22	81	30:70
9	$(CH_3)_2CHCH_2CHO$	22	76	30:70
10	*c*-C_6H_{11}CHO	22	83	10:90
11	Benzaldehyde	4	80	70:30

[a] External bath temperature ± 1 °C. [b] Reaction carried out with indium in H_2O; reaction time = 6 h. [c] *Syn*-**5** partially undergoes acetyl transfer (~ 10%) to 4-acetoxy-1-en-3-ol.

 DOI: 10.1039/b106759a

tion generally requires metalation of protected allyl ethers with alkyllithiums followed by transmetalation.



6a : RO = MeO, ML_n = $BIpc_2$
6b : RO = MeO, ML_n = $InCl_2$
6c : RO = MeO, ML_n = $FeCp(CO)_2$
6d : RO = TBSO, ML_n = $SnBu_3$
6e : RO = sugarO, ML_n = $SnBu_3$
6f : RO = sugarO, ML_n = $AlEt_3$

Indium may replace zinc (Table 1, run 1) but the overall process is slower and does not produce a significant improvement both in terms of yield and selectivity. With indium a Grignard protocol in THF seems to represent a more appropriate reaction system and results will be published independently.

Even though the stereochemical outcome of these reactions, in terms of simple diastereoselectivity, is not exceptional, an interesting difference comes out with respect to the use of **6a–f**. In fact, all the **6a–f** species are denoted by a remarkable *syn*-selectivity, independent of the aldehyde used, with *syn–anti* ratios in the 70 : 30–99 : 1 range. Invariably, the stereopreference for the formation of the *syn*-adduct is discussed in terms of a preferred formation of a *Z*-configurated organometallic species **6**, stabilised by internal coordination, which adds to the aldehyde following the classical chair-like Zimmerman-type transition state.

Results reported in Table 1 show that, starting from **3**, the simple diastereopreference depends on the nature of the aldehyde, saturated aldehydes mainly affording *anti* adducts (runs 6–10), and aromatic or α,β-unsaturated aldehydes leading to *syn* adducts (runs 1–5). No attempt was made to detect the actual structure of the intermediate **4**, anyway, in analogy to **6**, we believe that (*Z*)-**4** should enjoy thermodynamic stabilisation by zinc–oxygen coordination and should represent the most abundant isomer in the reaction mixture. On a purely speculative basis, we believe that the most important contribution to the observed diastereoselectivity is given by twist–boat transition states (TSs) **A** and **B** (Fig. 1) rather than by Zimmerman chair-like TSs which could not enjoy the through space stabilising O $\rightarrow$ Zn interaction present in **A** and **B**. **A** holds the sterically demanding R group in the *anti* position relative to the acetoxy group, so leading to the *anti* adduct, while **B** eclipses the unsaturated substituent with the acetoxy group with a likely stabilisation by π–π stacking interaction.

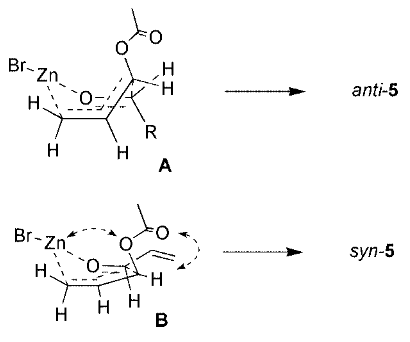


Fig. 1 Proposed transition states for the addition of **4** to aldehydes.

In conclusion, a new promising route to monoacetylated 1-en-3,4-diols based on 3-bromo-1-acetoxyprop-1-ene (**3**) is reported; this protocol joins an excellent efficiency in terms of yield, cost and time, to the use of typical environmentally benign conditions, such as the use of water as a solvent at room temperature.

This work was supported by MURST-Rome (National Project 'Stereoselezione in Sintesi Organica. Metodologie e Applicazioni') and University of Bologna (funds for selected topics).

Notes and references

† Anhydrous $ZnCl_2$ (0.068 g, 0.5 mmol) was added at -20 °C to a solution of freshly distilled propenal (3.32 mL, 50 mmol) and acetyl bromide (3.73 mL, 50 mmol) in anhydrous CH_2Cl_2 (40 mL). The reaction was stirred for 2 h at 0 °C and quenched at 0 °C by addition of aq. $NaHCO_3$. After extraction (CH_2Cl_2), 3-bromo-1-acetoxyprop-1-ene (**3**) was isolated by distillation (bp = 69–71 °C/22 mmHg) as a 35 : 65 mixture of *E*/*Z* isomers in 70% yield. (*Z*)-**3**. ^{1}H NMR (300 MHz, $CDCl_3$) δ 2.22 (s, 3H), 4.09 (dd, J = 0.3, 8.1 Hz, 2H), 5.24 (dt, J = 6.3, 8.1 Hz, 1H), 7.19 (dt, J = 0.3, 6.3 Hz, 1H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 20.5, 23.5, 109.3, 136.9, 166.9. (*E*)-**3.** ^{1}H NMR (300 MHz, $CDCl_3$) δ 2.16 (s, 3H), 3.99 (dd, J = 0.3, 8.1 Hz, 2H), 5.70 (dt, J = 8.1, 12.3 Hz, 1H), 7.19 (dt, J = 0.3, 12.3 Hz, 1H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 20.5, 28.4, 111.1, 139.0, 167.2.

‡ In a typical experimental procedure, an aldehyde (1 mmol) and **3** (1.5 mmol) are added to a sat aq. solution of NH_4Cl (5 ml). Zinc powder (1.5 mmol) is added and completely dissolves within 2 min. The reaction is checked after 15 min and products are extracted with ether after 30 min. The monoacetylated diol **5** is isolated by flash-chromatography in the 75–90% yield range. All new compounds were fully characterised by ^{1}H NMR, ^{13}C NMR and high resolution mass spectrometry; *syn–anti* stereorelationships were assigned after hydrolysis to 1-en-3,4-diols (Ref. 12).

1 P. T. Anastas and J. C. Warner, *Green Chemistry: Theory and Practice*, Oxford University Press, New York, 1998.
2 R. Ballini and G. Bosica, *J. Org. Chem.*, 1997, **62**, 425.
3 H. H. Baer, *Adv. Carbohydrate Chem. Biochem.*, 1969, **24**, 67.
4 A. Lubineau, J. Augé and Y. Queneau, *Synthesis*, 1994, 741; C. H. Li and T. H. Chan, *Tetrahedron*, 1999, **55**, 11149.
5 C. J. Li, *Tetrahedron*, 1996, **52**, 5643.
6 C. J. Li and W. C. Zhang, *J. Am. Chem. Soc.*, 1998, **120**, 9102.
7 E. Kim, D. M. Gordon, W. Schmid and G. M. Whitesides, *J. Org. Chem.*, 1993, **58**, 5500.
8 T. H. Chan and M. B. Isaac, *Pure Appl. Chem.*, 1996, **68**, 919.
9 Y. Canac, E. Levoirier and A. Lubineau, *J. Org. Chem.*, 2001, **66**, 3206.
10 A. R. Katritzky, M. Piffl, H. Lang and E. Anders, *Chem. Rev.*, 1999, **99**, 665.
11 M. Neuenschwander, P. Bigler, K. Christen, R. Iseli, R. Kyburz and H. Mühle, *Helv. Chim. Acta*, 1978, **61**, 2047.
12 **6a**: P. Ganesh and K. M. Nicholas, *J. Org. Chem.*, 1997, **62**, 1737; **6b**; T. Hirashita, T. Kamei, T. Horie, H. Yamamura, M. Kaway and S. Araki, *J. Org. Chem.*, 1999, **64**, 172; **6c**; S. Jiang, G. E. Agoston, T. Chen, M. P. Cabal and E. Turos, *Organometallics*, 1995, **14**, 4697; **6d**; G. E. Keck, D. E. Abbott and M. R. Wiley, *Tetrahedron Lett.*, 1987, **28**, 139; **6e**; W. R. Roush and M. S. VanNieuwenhze, *J. Am. Chem. Soc.*, 1994, **116**, 8536; **6f**; J.-I. Chika and H. Takei, *Tetrahedron Lett.*, 1998, **39**, 605.